ProctorAI

AI-assisted complaint, identity & access platform

OPERATIONS

Configuration Reference

Every environment variable, default, owner and effect across the ProctorAI stack.

DOCUMENT ID	PRODUCT VERSION	DOCUMENT VERSION	ISSUED	CLASSIFICATION
PAI-CFG-007	1.1.0	1.0	November 2026	Confidential

CREATOR & DEVELOPER

Varshit Tyagi & Mohini Teotia

Document control

Title	Configuration Reference
Document ID	PAI-CFG-007
Audience	Operators and engineers
Product	ProctorAI 1.1.0
Status	Approved for use
Classification	Confidential

Creator & Developer

Name	Role
Varshit Tyagi	Creator & Developer
Mohini Teotia	Creator & Developer

Revision history

Version	Date	Author	Change
1.0	November 2026	Varshit Tyagi, Mohini Teotia	Initial release for ProctorAI 1.1.0

Related documents

Deployment Runbook (PAI-DEP-006), Security & Hardening (PAI-SEC-005)

© 2026 Varshit Tyagi & Mohini Teotia. ProctorAI is released under the MIT License. Product names mentioned are trademarks of their respective owners.

Contents

1 Configuration model
1.1 Ownership legend
2 Database
3 Authentication and sessions
4 Gateway, web and CORS
5 Domains and TLS (production)
6 Service registry
7 Redis
8 Object storage (MinIO / S3)
9 Temporary files
10 Notifications
11 Face recognition
12 Versions and images
13 Frontend and native apps
14 JVM and container settings (compose)
15 Ports
16 Change procedure

1. Configuration model

ProctorAI is configured entirely through environment variables. Setup scripts generate one env file per stack:

Stack	File	Created by
Development	`backend/.env`	`./setup.sh`
Production from source	`backend/.env.production`	`./setup.sh prod`
Production from published images	`docker/.env`	`docker/setup.sh`

All env files are created with mode `600`. `backend/.env.example` lists every variable with placeholders.

1.1 Ownership legend

Owner	Meaning
Setup (once)	Generated on first run and never changed automatically. Changing it later requires a coordinated rotation.
Setup (every run)	Rewritten on every setup run from what is actually served. Do not edit by hand.
Prompt	Asked by setup; rerun setup to change.
You	Safe to edit; rerun setup (`--yes`) to apply.
Compose override †	Read by the service, but not passed through by any compose file in 1.1.0. Setting it in the env file has no effect. To change it, add it to the service's `environment:` block in the compose file used by setup, then rerun setup.

> **IMPORTANT**
>
> After editing an env file, apply it with `./setup.sh` (development), `./setup.sh prod --yes` (source) or `./setup.sh` in `docker/` (published images). The production compose file fails fast when a required variable is missing.

2. Database

Variable	Default / generated	Owner	Description
`POSTGRES_DB`	`proctor`	Setup (once)	Database name
`POSTGRES_USER`	`proctor_admin`	Setup (once)	Database owner; used only by `db-migrator`
`POSTGRES_PASSWORD`	32 random chars	Setup (once)	Owner password

Variable	Default / generated	Owner	Description
`AUTHENTICATOR_USER_NAME`	`authenticator`	Setup (once)	Role used by auth-service
`AUTHENTICATOR_USER_PASSWORD`	32 random chars	Setup (once)	Password of `authenticator` (set by the migrator)
`APP_USER_NAME`	`proctor`	Setup (once)	Role used by all other services
`APP_USER_PASSWORD`	32 random chars	Setup (once)	Password of `proctor` (set by the migrator)
`ADMIN_PASSWORD`	`admin123` (dev) / 20 random chars (prod)	Setup (once)	Password of `admin@college.com` on a new database
`R2DBC_URL`	`r2dbc:postgresql://postgres:5432/<db>` in containers	Compose	Reactive connection URL
`SPRING_DATASOURCE_URL`	`jdbc:postgresql://postgres:5432/<db>` (face-service)	Compose	JDBC URL for Spring AI vector store
`MIGRATION_DATABASE_URL`	`jdbc:postgresql://postgres:5432/<db>`	Compose	JDBC URL for Flyway
`POSTGRES_PORT`	`5433`	You (dev)	Host port for PostgreSQL in development

3. Authentication and sessions

Variable	Default	Owner	Description
`JWT_SECRET`	64 random chars	Setup (once)	HMAC key for access tokens; must be ≥ 32 bytes
`JWT_ISSUER`	`proctor.local`	Setup (once)	`iss` claim
`JWT_ACCESS_EXPIRE_IN_SEC`	`900`	You	Access token lifetime (15 minutes)
`JWT_REFRESH_EXPIRE_IN_SEC`	`2592000`	You	Refresh token lifetime (30 days)
`APP_COOKIE_SECURE`	`false` (dev) / `true` (prod)	Setup (every run)	`Secure` flag on cookies
`APP_COOKIE_SAME_SITE`	`Lax`	Compose override †	`SameSite` attribute of cookies
`LOGIN_MAX_FAILURES`	`10`	You	Failed sign-ins before an account is locked
`LOGIN_LOCKOUT`	`15m`	You	Lockout duration (Spring duration format: `30s`, `15m`, `1h`)

4. Gateway, web and CORS

Variable	Default	Owner	Description
`APP_ALLOWED_ORIGINS`	Generated	Setup (every run)	CORS origins: dev `http://localhost:<port>`, `http://127.0.0.1:<port>`; prod `https://<app domain>` (or `http://<ip>` before DNS) plus native app origins
`APP_EXTRA_ORIGINS`	empty	You	Extra allowed origins, comma separated. Validated: no wildcards, HTTP(S) only, storage and registry origins refused
`APP_TRUSTED_PROXIES`	`1` (dev) / `2` (prod)	Compose	Number of proxies that append to `X-Forwarded-For` (Caddy and Nginx in production)
`LOGIN_RATE_LIMIT_REPLENISH`	`1`	You	Tokens added per second to each client IP bucket
`LOGIN_RATE_LIMIT_BURST`	`30`	You	Bucket capacity
`LOGIN_RATE_LIMIT_COST`	`3`	You	Tokens per sign-in attempt (defaults: 10 immediate attempts, then 20/min)
`GATEWAY_SERVER_PORT`	`7050`	You (dev)	Host port of the gateway in development
`FRONTEND_PORT`	`5173`	You (dev)	Host port of the web app in development

5. Domains and TLS (production)

Variable	Example	Owner	Description
`APP_DOMAIN`	`app.example.com`	Prompt	Web app and API
`FILES_DOMAIN`	`files.example.com`	Prompt	Object storage for presigned URLs
`REGISTRY_DOMAIN`	`registry.example.com`	Prompt	Eureka dashboard
`ACME_EMAIL`	`ops@example.com`	Prompt	Let's Encrypt account email
`PUBLIC_IP`	detected	Prompt / auto	Server public IPv4, used before DNS is ready

6. Service registry

Variable	Default	Owner	Description
`DISCOVERY_USERNAME`	`proctor-discovery`	Setup (once)	Basic auth user for Eureka
`DISCOVERY_PASSWORD`	48 random chars	Setup (once)	Basic auth password (≥ 32 chars, enforced)

Variable	Default	Owner	Description
`DISCOVERY_SERVER_PORT`	`8761`	You	Host port (bound to `127.0.0.1` in production)
`DISCOVERY_SERVER_REFRESH_INTERVAL`	`2`	You (dev)	Registry refresh interval in seconds
`DISCOVERY_DASHBOARD_ENABLED`	`true` in prod compose	Compose	Serves the dashboard behind the registry domain
`EUREKA_CLIENT_SERVICEURL_DEFAULTZONE`	Built from the credentials	Compose	Registry URL used by every service

7. Redis

Variable	Default	Owner	Description
`REDIS_HOST`	`redis` in containers	Compose	Host
`REDIS_PORT`	`6379` in containers / `6390` dev host port	Compose / You	Port
`REDIS_PASSWORD`	32 random chars	Setup (once)	`requirepass` password

8. Object storage (MinIO / S3)

Variable	Default	Owner	Description
`MINIO_ROOT_USER`	`proctorminio`	Setup (once)	MinIO root user (console and init only)
`MINIO_ROOT_PASSWORD`	32 random chars	Setup (once)	MinIO root password
`S3_ACCESS_KEY`	`proctor-app`	Setup (once)	Application user, limited to one bucket
`S3_SECRET_KEY`	40 random chars	Setup (once)	Application user secret
`S3_BUCKET`	`proctor-files`	Setup (once)	Bucket for complaint files
`S3_ENDPOINT`	`http://minio:9000` in containers	Compose	Internal S3 endpoint
`S3_PUBLIC_ENDPOINT`	`http://localhost:9200` (dev) / `https://<files domain>`	Setup (every run)	Base URL in presigned links; must be reachable by clients
`MINIO_CORS_ALLOW_ORIGIN`	App origin	Setup (every run)	Origins allowed to upload directly to MinIO
`S3_PORT`	`9200`	You	Host port for the S3 API (dev; prod only before DNS)
`S3_CONSOLE_PORT`	`9201`	You	Host port of the MinIO console (bound to `127.0.0.1`)

9. Temporary files

Variable	Default	Owner	Description
`APP_STORAGE_DIR`	`./storage` (dev) / `/data` (containers)	Compose	Directory for import job files (user-service, face-service)
`APP_STORAGE_TEMP_TTL`	`30m`	Compose override †	Maximum age of temporary job files
`APP_STORAGE_SWEEP_INTERVAL`	`15m`	Compose override †	How often the sweeper runs

10. Notifications

Variable	Default	Owner	Description
`NOTIFICATION_SECRET_KEY`	48 random chars	Setup (once)	Encrypts FCM service-account secrets (AES-256-GCM). Rotating it requires re-entering FCM credentials

FCM credentials themselves are entered in the application (Admin → Notifications), not in env files.

11. Face recognition

Variable	Default	Owner	Description
`FACE_SERVER_PORT`	`7059`	Compose	HTTP port
`FACE_DETECTION_MODEL_PATH`	`./models/face-detector.onnx` (image: `/app/models/…`)	Image	SCRFD model file
`FACE_EMBEDDING_MODEL_PATH`	`./models/face-embedder.onnx`	Image	ArcFace model file
`FACE_DETECTION_INPUT_SIZE`	`640`	You	Detector input square size
`FACE_DETECTION_SCORE_THRESHOLD`	`0.5`	You	Minimum detector confidence
`FACE_DETECTION_NMS_THRESHOLD`	`0.4`	You	Non-maximum suppression IoU
`FACE_MATCH_THRESHOLD`	`0.45` in generated env files	You	Minimum cosine similarity to report a match
`FACE_MAX_IMAGE_BYTES`	`8388608`	You	Maximum decoded image size (8 MB)
`FACE_EMBEDDING_DIMENSIONS`	`512`	Image	Must match the embedder output

Variable	Default	Owner	Description
`FACE_SELF_ENROLL_LIMIT`	`1`	Compose override †	Self-enrollment attempts before lock

> **NOTE**
> Generated env files set `FACE_MATCH_THRESHOLD=0.45`. The production compose file falls back to `0.75` only when the variable is absent. Higher values reduce false matches but increase missed matches. Validate the threshold on a sample of your own users before go-live.

12. Versions and images

Variable	Default	Stack	Description
`APP_VERSION`	`latest`	Source production	Tag for locally built images
`PROCTOR_VERSION`	`1.1.0`	Published images	Image version pulled from Docker Hub (`./setup.sh update <version>` sets it)
`IMAGE_REPOSITORY`	`vishaltyagi807/proctor-ai`	Published images	Docker Hub repository

13. Frontend and native apps

Variable	Where	Description
`VITE_SERVER_URL`	Build time (`npm run build`, `tauri:*:build`)	Server URL baked into native apps; when empty, the app asks at sign-in
`APPLE_DEVELOPMENT_TEAM`	iOS builds	Apple team id for signing
`APPLE_SIGNING_IDENTITY`, `APPLE_ID`, `APPLE_PASSWORD`, `APPLE_TEAM_ID`	macOS builds	Code signing and notarisation
`ANDROID_HOME`, `NDK_HOME`, `JAVA_HOME`	Android builds	Toolchain locations

14. JVM and container settings (compose)

Setting	Development	Production
`JAVA_TOOL_OPTIONS`	C1 only, Serial GC, CDS archive (fast start)	`-XX:+UseG1GC -XX:MaxRAMPercentage=60 -XX:InitialRAMPercentage=25 -XX:+ExitOnOutOfMemoryError`

Setting	Development	Production
Memory limits	None	768 MB per service, 2560 MB face-service, 512 MB registry and monitor, 384 MB migrator, 256 MB edge, 128 MB frontend
Health check	`/actuator/health`	Same, 15 s interval, 120 s start period
Logging	Docker default	`json-file`, 10 MB × 5

15. Ports

Port	Development (host)	Production (host)
80 / 443 tcp, 443 udp	—	Caddy edge (public)
5173	Web app	—
7050	Gateway	—
7051–7060	Each service on `127.0.0.1`	—
8761	Registry	`127.0.0.1` only
5433	PostgreSQL	—
6390	Redis	—
9200	MinIO API	Public only before DNS
9201	MinIO console	`127.0.0.1` only

16. Change procedure

1. Back up the env file: `cp .env .env.bak-$(date +%F)` (keep permissions `600`).
2. Edit only variables owned by **You** or **Prompt**.
3. Apply with the setup command for the stack.
4. Check `./setup.sh status` and the affected service logs.
5. Record the change in the operations log.